Aviraj Patil — Resume

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Contact

Full-stack developer building web apps, mobile apps and local AI tools

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Summary

Second-year B.Tech CSE (AI & ML) student at JSPM, Pune. I build complete products, from the database and backend to the UI, payments and deployment. I work mostly in TypeScript (Next.js, Expo, Supabase) and Python (FastAPI). I also build AI pipelines that run on local hardware. I have shipped 25+ projects, including a calorie-tracking app for Indian food, an automated Hindi cartoon studio, and three hackathon builds.

Projects

Katori: calorie and fitness app for Indian food. Expo, TypeScript, Supabase, Deno

- Photo-based meal logging: the AI names the dishes and portions, and the server computes macros from a reviewed Indian food table.
- Row-level security on every table, with automated database tests. CI applies migrations and deploys edge functions and the PWA on every push.
- Razorpay subscriptions, a trainer mode with per-client billing, an offline workout logger and a Telegram bot.

GramToon Studio: automated Hindi cartoon studio. Python, pycairo, Ollama, ComfyUI, n8n, FFmpeg

- Turns an idea into a 1080p animated episode: story, rigged characters, voices with lip-sync, animation, sound mix, render and a 9-check QC.
- Renders 74 seconds of video in 72 seconds on a 6 GB laptop GPU, with no paid APIs.

PODSYU: podcast-to-Shorts generator. FastAPI, faster-whisper, Ollama, Next.js

- Finds self-contained moments in long videos, reframes them to 9:16 by tracking the speaker, and burns in word-synced captions.
- Publishes and schedules to YouTube through the YouTube Data API.

Sukyoo: app for Indian senior citizens and their families. Expo, Next.js, Supabase

- Senior app, family app and web app in one monorepo, covering healthcare, shopping and events.

MedFlow: blood test pre-analysis for hospitals. FastAPI, Next.js, Groq

- Six AI agents read a patient's blood report before the consultation, so the doctor starts with a summary.

UNMATRIX Store: digital products store. Next.js, Postgres, Cashfree

- Accounts, UPI checkout, delivery links and an admin dashboard with orders, customers and analytics.

Hackathons

Smart India Hackathon 2026: **VASP Trace** (problem statement SIH26182, MHA / I4C). Python, NetworkX, Streamlit, Postgres

- Attributes unknown crypto wallets to the exchange they most likely belong to: an Ethereum trace pipeline, exchange clustering by hot-wallet sweeps, and mixer and bridge typologies.

PSBs Hackathon Series 2026 (IDEA 2.0): **ARIA**. FastAPI, Next.js

- AI banking assistant for public sector banks across three channels: portal chat, voice bot and SMS bot, sharing one backend.

Meta OpenEnv hackathon: **MisInfo Arena** **and** **Peer Review Arena**. Python, Docker

- Reinforcement learning environments where an LLM agent learns to fact-check claims, sources and coordinated cascades, across three difficulty levels.

Websites for clients

- **Jayesh Kand Social Foundation**: bilingual Marathi and English site for an NGO near Pune. Next.js
- **dr.da labs**: fragrance science site for a chemistry professor at JSPM University. HTML, CSS, JavaScript
- **NextGenTech**: company site with solutions, research, projects and campus ambassador pages
- **Interior Studio** and an operations system for interior designers. TypeScript
- **MGM student ERP**: student portal UI for attendance, timetable, fees, results and library

Skills

- **Languages:** TypeScript, JavaScript, Python, SQL, HTML, CSS
- **Frontend and mobile:** React, Next.js, Expo / React Native, Tailwind CSS, Vite
- **Backend and data:** FastAPI, Node.js / Express, Supabase (Postgres, Auth, Row-Level Security, Edge Functions), Deno
- **AI and media:** LLM APIs, Ollama, faster-whisper, ComfyUI / Stable Diffusion, ONNX TTS, FFmpeg, Blender scripting
- **DevOps and tools:** Git, GitHub Actions, Vercel, Cloudflare Pages, Docker, n8n, Vitest, pytest
- **Payments:** Razorpay, Cashfree (UPI)

Education

B.Tech, Computer Science and Engineering (AI & ML), JSPM, Wagholi, Pune. 2025 to 2029 (currently in second year)